

Algebraic Groups

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Abstract

References:

- Fourier-Mukai Transforms in Algebraic Geometry (Derived Categories for Algebraic Geometry)

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1 Introduction to Group Schemes

1.1 Group Schemes in General

Definition 1.1.1 (Group Schemes)

Let S be a scheme. A group scheme over S is a group object in $\mathbf{Sch}_S$.

Definition 1.1.2 (Types of Group Schemes)

Let S be a scheme. Let G be a group scheme over S .

- We say that G is a group variety if G is a variety.
- We say that G is an affine group scheme if G is affine.
- We say that G is an abelian variety if G is a complete connected group variety.

Let X be an S -scheme. Recall that the functor of points is the functor $h_X : \mathbf{Sch}_S \rightarrow \mathbf{Set}$ given by

$$h_X(Y) = X(Y) = \mathrm{Hom}_{\mathbf{Sch}_S}(Y, X)$$

Lemma 1.1.3

Let S be a scheme. Let G be a scheme over S . Then G is a group scheme if and only if its functor of points $h_G : \mathbf{Sch}_S \rightarrow \mathbf{Set}$ factors through the forgetful functor $\mathbf{Grp} \rightarrow \mathbf{Set}$.

Definition 1.1.4 (Commutative Group Schemes)

Let S be a scheme. A commutative group scheme over S is an abelian group object in $\mathbf{Sch}_S$.

Lemma 1.1.5

Let S be a scheme. Let G be a group scheme over S . Then the following are equivalent.

- G is commutative.
- $G(X)$ is an abelian group for all $X \in \mathbf{Sch}_S$.
- The functor of points $h_G : \mathbf{Sch}_S \rightarrow \mathbf{Set}$ factors through the forgetful functor $\mathbf{Ab} \rightarrow \mathbf{Grp}$.

Definition 1.1.6 (Morphism of Group Schemes)

Let S be a scheme. Let G, H be group schemes over S . Let $\varphi : G \rightarrow H$ be a morphism over S . We say that φ is a morphism of group schemes if it is a morphism of group objects in $\mathbf{Sch}_S$.

Equivalently, this means that φ must respect multiplication:

$$\varphi \circ m_G = m_H \circ (\varphi \times \varphi)$$

Lemma 1.1.7

Let S be a scheme. Let G, H be group schemes over S . Let $\varphi : G \rightarrow H$ be a morphism over S . Then φ is a morphism of group schemes if and only if $h_\varphi(X) : h_G(X) \rightarrow h_H(X)$ is a group homomorphism for all $X \in \mathbf{Sch}_S$.

Definition 1.1.8 (Group Subschemes)

Let S be a scheme. Let G be a group scheme over S . A group subscheme of G is a subset $H \subseteq G$ such that the following are true.

- H is a subscheme of G .
- The inclusion $H \hookrightarrow G$ is a morphism of group schemes.

Lemma 1.1.9

Let S be a scheme. Let G, H be group schemes over S . Then H is a group subscheme of G if and only if $h_H(X)$ is a subgroup of $h_G(X)$ for all $X \in \mathbf{Sch}_S$.

1.2 Group Schemes of Finite Type Over a Finitely Generated Algebra

Let k be a field. Let R be a finitely generated commutative k -algebra. Let X be a scheme of finite type over R . We have to be careful about the conditions on the functor of points for X to be a group scheme. There are two functor of points $h_X : \mathbf{Sch}_R \rightarrow \mathbf{Set}$ and $h_X^{\text{ft}} : \mathbf{Sch}_R^{\text{ft}} \rightarrow \mathbf{Set}$. Normally we use the former functor to check if X is a group scheme no matter what conditions we impose on X . We need two facts:

1. $h_X : \mathbf{Sch}_R \rightarrow \mathbf{Set}$ commutes with filtered colimits of commutative R -algebras / R -schemes.
2. Every R -scheme is a filtered colimit of its finite type pieces.

But we can do better. Since X is a finite type scheme, it is covered by finitely many affine finite type schemes. Fact: h_X can be reconstructed from h_U where U ranges over the affine finite type schemes of X . Thus it suffices to check h_X over all affine finite type schemes.

Finally, using the equivalence between finite generated R -algebras and affine finite type schemes over R , we have the following:

Proposition 1.2.1

Let k be a field. Let R be a finitely generated commutative k -algebra. Let X be a scheme of finite type over R . Then X is a group scheme if and only if the functor $\tilde{X} : \mathbf{FGCAlg}_R \rightarrow \mathbf{Set}$ given by

$$A \mapsto \text{Hom}_{\mathbf{Sch}_R}(\text{Spec}(A), X)$$

factors through the forgetful functor $\mathbf{Grp} \rightarrow \mathbf{Set}$.

Note that Milnor book does not go through this because he defines group schemes as group objects directly in the category of schemes of finite type over R .

The checks for commutativity, subgroup and homomorphism also follows the same reduction.

Lemma 1.2.2

Let k be a field. Let R be a finitely generated commutative k -algebra. Let G be a group scheme of finite type over R . Then the following are equivalent.

- G is commutative.
- $\tilde{G}(A)$ is an abelian group for all $A \in \mathbf{FGCAlg}_R$.
- $\tilde{G}$ factors through the forgetful functor $\mathbf{Ab} \rightarrow \mathbf{Grp}$.

Lemma 1.2.3

Let k be a field. Let R be a finitely generated commutative k -algebra. Let G, H be group schemes of finite type over R . Then the following are true.

- Let $\varphi : G \rightarrow H$ be a morphism of schemes. Then φ is a morphism of group schemes if and only if $\tilde{\varphi}(A)$ is a group homomorphism for all $A \in \mathbf{FGCAlg}_R$.
- H is a group subscheme of G if and only if $\tilde{H}(A)$ is a subgroup of $\tilde{G}(A)$ for all $A \in \mathbf{FGCAlg}_R$.

Example 1.2.4

Let k be a field.

- The functor $A \mapsto (A, +)$ is represented by the group scheme

$$\mathbb{G}_a(k) = \text{Spec}(k[x])$$

- The functor $A \mapsto (A^\times, \times)$ is represented by the group scheme

$$\mathbb{G}_m(k) = \text{Spec}(k[x, x^{-1}])$$

- Let $n \in \mathbb{N}$. The functor $A \mapsto \{a \in A \mid a^n = 1\}$ is represented by the group scheme

$$\mu_n = \operatorname{Spec} \left(\frac{k[x]}{(x^n - 1)} \right)$$

- The functor $A \mapsto \operatorname{GL}(n, A)$ is represented by the group scheme

$$\operatorname{GL}(n, k) = \operatorname{Spec} \left(\frac{k[x_{i,j} \mid 1 \leq i, j \leq n] \otimes_k k[y]}{(\det((x_{i,j})_{1 \leq i, j \leq n})y - 1)} \right)$$

Example 1.2.5

Let k be a field.

- Let V be a finite dimensional vector space over k . Then the functor $A \mapsto V \otimes_k A$ is represented by the group scheme

$$V_a = \operatorname{Spec}(S(V^*))$$

- Let V be a finite dimensional vector space over k . Let $\phi : V \rightarrow k^n$ be a k -isomorphism. Then ϕ induces an isomorphism

$$V_a \cong \mathbb{G}_a^{\dim(V)}$$

- Let V be a finite dimensional vector space over k . Define the group scheme $\operatorname{GL}(V)$ to be the representing object of the functor $A \mapsto \operatorname{Aut}_k(V \otimes_k A)$.
- Let V be a finite dimensional vector space over k . Let $\phi : V \rightarrow k^n$ be a k -isomorphism. Then the map $T \mapsto \phi \circ T \circ \phi^{-1}$ induces an isomorphism of group schemes

$$\operatorname{GL}(V) \cong \operatorname{GL}(n, k)$$

1.3 Dense Subgroups

Definition 1.3.1 (Schematically Dense)

Let k be a field. Let X be a scheme of finite type over k . Let S be a subset of $X(k)$. We say that S is schematically dense in X if the map $\mathcal{O}_X(X) \rightarrow \prod_{s \in S} k(s)$ defined by $f \mapsto (f(s))_{s \in S}$ is injective.

Proposition 1.3.2

Let k be a field. Let G be a group scheme over k . Then the following are equivalent.

- $G(k)$ is dense in G .
- $G(k)$ is schematically dense in G .
- G is reduced and $G(k)$ is dense in the subspace of closed points of G .

1.4 The Identity Component

Lemma 1.4.1

Let k be a field. Let G be a group scheme of finite type over k . Then the following are true.

- G is irreducible if and only if G is connected.
- G is the disjoint union of its irreducible components.

Definition 1.4.2 (Identity Component)

Let k be a field. Let G be a group scheme of finite type over k . Define the identity component G° of G to be the unique irreducible component of G that contains 1_G .

Proposition 1.4.3

Let k be a field. Let G be a group scheme of finite type over k . Then G° is a subgroup scheme of G .

Proposition 1.4.4

Let k be a field. Let G be a group scheme of finite type over k . Let F/k be a field extension. Then we have

$$G^\circ(F) \cong G(F)^\circ$$

1.5 Reducedness and Smoothness**Proposition 1.5.1**

Let k be a field. Let G be a group scheme of finite type over k . Then G is separated.

Proposition 1.5.2

Let k be a field. Let G be a group scheme of finite type over k . Then the following are equivalent.

- G is geometrically reduced.
- G is smooth.
- G° is smooth.
- 1_G is a smooth point of G .
- $\dim(T_1 G) = \dim(G)$.

Proposition 1.5.3

Let k be a field. Let G be a group scheme of finite type over k . If G is reduced and k is perfect, then G is geometrically reduced.

Thus group varieties over a perfect field are smooth.

Proposition 1.5.4

Let k be a field. Let G be a group scheme of finite type over k . Then the following are equivalent.

- G is irreducible.
- G is connected.
- G is geometrically connected.

1.6 Correspondence of Subgroup Schemes**Proposition 1.6.1**

Let k be a field. Let G be a group scheme of finite type over k . If G_{red} is geometrically reduced, then it is a subgroup scheme.

Proposition 1.6.2

Let k be a field. Let G be a group scheme of finite type over k . Let $H \leq G$ be a subgroup scheme of G . Then H is closed in G .

Thus subgroup schemes of affine group schemes are affine.

Proposition 1.6.3

Let k be a field. Let G be a group scheme of finite type over k . Then there is a one to one correspondence

$$\left\{ S \leq G(k) \mid S \text{ is a closed subgroup} \right\} \xleftrightarrow{1:1} \left\{ H \subseteq G \mid \begin{array}{l} H \text{ is a geometrically reduced} \\ \text{subgroup scheme such that} \\ H(k) \text{ is schematically dense in } H \end{array} \right\}$$

given by $H \mapsto H(k)$ in the backwards direction.

Since geometrically reduced implies reduced and all group schemes are separated, we deduce the following corollary.

Corollary 1.6.4

Let k be a field. Let G be a group scheme of finite type over k . Then there is a one to one correspondence

$$\left\{ S \leq G(k) \mid S \text{ is a closed subgroup} \right\} \xleftrightarrow{1:1} \left\{ H \subseteq G \mid \begin{array}{l} H \text{ is a subgroup variety of } G \text{ such} \\ \text{that } H(k) \text{ is topologically dense in } G \end{array} \right\}$$

given by $H \mapsto H(k)$ in the backwards direction.

Corollary 1.6.5

Let k be a separable field. Let G be a group scheme of finite type over k . Then there is a one to one correspondence

$$\left\{ S \leq G(k) \mid S \text{ is a closed subgroup} \right\} \xleftrightarrow{1:1} \left\{ H \subseteq G \mid H \text{ is a subgroup variety of } G \right\}$$

given by $H \mapsto H(k)$ in the backwards direction.

1.7 The Affine Case and Hopf Algebras

Proposition 1.7.1

Let k be a field. Let A be a finitely generated commutative algebra over k . Let $\Delta : A \rightarrow A \otimes_k A$ be a k -algebra homomorphism. Then $(\text{Spec}(A), \text{Spec}(\Delta))$ is an affine group scheme over k if and only if (A, Δ) is a commutative Hopf algebra.

Proposition 1.7.2

Let k be a field. There is an equivalence of categories

$$\text{Spec} : \text{FGCHopf}_k \xrightarrow{\cong} \text{AffGrpSch}_k$$

Proposition 1.7.3

Let k be a field. Let G be an affine group scheme of finite type over k . Then the correspondence between closed subschemes of G and ideal of $\mathcal{O}_G(G)$ descends to a one to one correspondence

$$\left\{ H \leq G \mid H \text{ is a subgroup scheme of } G \right\} \xleftrightarrow{1:1} \left\{ I \subseteq \mathcal{O}_G(G) \mid I \text{ is a Hopf ideal} \right\}$$

Theorem 1.7.4 (Cartier 1962)

Let k be a field of characteristic 0. Let G be an affine group scheme of finite type over k . Then G is smooth.

1.8 Anti-Affine Groups

Definition 1.8.1 (Anti-Affine Groups)

Let k be a field. Let G be a group scheme of finite type over k . We say that G is anti-affine if $\mathcal{O}_G(G) = k$.

Lemma 1.8.2

Let k be a field. Let G be an abelian variety over k . Then G is anti-affine.

Proposition 1.8.3

Let k be a field. Let G be a group scheme of finite type over k . If G is anti-affine, then G is smooth and connected.

2 Group Actions and Representations

2.1 Group Actions

Definition 2.1.1 (Group Actions)

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . A group action of G on X is a morphism $\mu : G \times X \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \text{id}_X} & G \times X \\ \text{id}_G \times \mu \downarrow & & \downarrow \mu \\ G \times X & \xrightarrow{\mu} & X \end{array} \quad \begin{array}{ccc} * \times X & \xrightarrow{\text{incl.}} & G \times X \\ & \searrow \cong & \downarrow \mu \\ & & X \end{array}$$

Lemma 2.1.2

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a morphism. Then μ is a group action if and only if $\tilde{\mu} : \tilde{G} \times \tilde{X} \rightarrow \tilde{X}$ is a natural transformation such that for all $R \in \mathbf{FAlg}_k$, $\mu(R) : G(R) \times X(R) \rightarrow X(R)$ is a group action.

Lemma 2.1.3

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a group action. Then μ is faithfully flat.

Lemma 2.1.4

Let k be a field. Let G be a group scheme over k . Let X be a scheme of finite type over k . Let $\mu : G \times X \rightarrow X$ be a group action. Then the following are true.

- If G is smooth, then μ is smooth.
- If G is finite, then μ is finite.

2.2 Representations of Affine Group Schemes

Definition 2.2.1 (Representation of a Group Scheme)

Let k be a field. Let G be a group scheme of finite type over k . A representation of G is a vector space V over k together with a morphism of group schemes $\rho : G \rightarrow GL(V)$.

We will see that for any group scheme H , there is a natural isomorphism

$$\text{Hom}_{\mathbf{GrpSch}_k}(G, H) \cong \text{Hom}_{\mathbf{GrpSch}_k}(G^{\text{aff}}, H)$$

so it suffices to consider affine group schemes.

Lemma 2.2.2

Let k be a field. Let G be an affine group scheme over k . Let V be a finite dimensional vector space over k . If $\rho : G \rightarrow GL(V)$ is a natural transformation, then ρ is a homomorphism of group schemes.

Lemma 2.2.3

Let k be a field. Let G be an affine group scheme over k . Let V be a finite dimensional vector space over k . Then there is a one to one correspondence

$$\{\rho : G \rightarrow GL(V) \mid \rho \text{ is a representation}\} \xrightarrow{1:1} \left\{ \phi : G \times V_a \rightarrow V_a \mid \begin{array}{l} \phi \text{ is a group action such that the action} \\ G(R) \times V_a(R) \rightarrow V_a(R) \text{ is } R\text{-linear} \end{array} \right\}$$

Proposition 2.2.4

Let k be a field. Let G be an affine group scheme over k . Let V be a finite dimensional vector space

over k . Then there is a one to one correspondence

$$\{\rho : G \rightarrow GL(V) \mid \rho \text{ is a representation}\} \xleftrightarrow{1:1} \left\{ \phi : V \rightarrow V \otimes \mathcal{O}_G(G) \mid \begin{array}{l} \phi \text{ defines a comodule} \\ \text{structure on } V \end{array} \right\}$$

Proposition 2.2.5

Let k be a field. Let G be an affine group scheme over k . Let V be a vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Then ρ is a filtered union of its finite dimensional subrepresentation.

Therefore it suffices to consider finite dimensional representations of affine group schemes.

2.3 The Regular Representation

Definition 2.3.1 (Faithful)

Let k be a field. Let G be a group scheme over k . Let V be a finite dimensional vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. We say that ρ is faithful if $\rho(R)$ is injective for all $R \in \mathbf{FAlg}_k$.

Definition 2.3.2 (The Regular Representation)

2.4 Linear Group Schemes

Definition 2.4.1 (Linear Group Schemes)

Let k be a field. Let G be a group scheme over k . We say that G is linear if there exists a finite dimensional vector space V over k together with a faithful representation $\rho : G \rightarrow GL(V)$.

Proposition 2.4.2

Let k be a field. Let G be a group scheme over k . Then G is a linear group scheme if and only if G is an affine group scheme.

Corollary 2.4.3

Let k be a field. Let G be an affine group scheme of finite type over k . Then G is isomorphic to a subgroup scheme of $GL(n, k)$ for some $n \in \mathbb{N}^+$.

2.5 Characters and Eigenspaces

Definition 2.5.1 (Characters)

Let k be a field. Let G be an group scheme of finite type over k . Define the group of characters of G to be

$$\chi(G) = \text{Hom}_{\mathbf{GrpSch}_k}^{\text{ft}}(G, \mathbb{G}_m(k))$$

Lemma 2.5.2

Let k be a field. Let G be an group scheme of finite type over k . Then $\chi(G)$ is an abelian group.

Proposition 2.5.3

Let k be a field. Let G be an group scheme of finite type over k . Then there is a bijection

$$\chi(G) \xleftrightarrow{1:1} G(\mathcal{O}_G(G))$$

where the latter is the group-like elements of the Hopf algebra.

We denote the above map by $a : \chi(G) \rightarrow G(\mathcal{O}_G(G))$.

Proposition 2.5.4

Let k be a field. Let G be a group scheme of finite type over k . Let $\chi_1, \dots, \chi_n \in \chi(G)$ be distinct characters. Then they are linearly independent as elements of the k -algebra $\mathcal{O}_G(G)$.

Definition 2.5.5 (Eigenspaces)

Let k be a field. Let G be a group scheme of finite type over k . Let V be a finite dimensional vector space over k . Let $\rho : G \rightarrow GL(V)$ be a representation. Let $\chi \in \chi(G)$ be a character. Define the eigenspace of χ by

$$V_\chi = \{v \in V \mid \rho(v) = v \otimes a(\chi)\}$$

3 The Category of Group Schemes

3.1 Kernels and Monomorphisms

Definition 3.1.1 (Kernels)

Proposition 3.1.2

Let k be a field. Let G, H be group schemes of finite type over k . Let $\varphi : G \rightarrow H$ be a morphism of group schemes. Then the following are equivalent.

- φ is a monomorphism in $\mathbf{GrpSch}_k^{\text{ft}}$.
- φ is a monomorphism in $\mathbf{Sch}_k^{\text{ft}}$.
- $\varphi(R) : G(R) \rightarrow H(R)$ for all $R \in \mathbf{FGCAlg}_k$.
- $\ker(\varphi) = \{1_G\}$.

3.2 Quotients and Epimorphisms

Definition 3.2.1 (Quotient Maps)

Let k be a field. Let G, H be group schemes of finite type over k . Let $\varphi : G \rightarrow H$ be a morphism of group schemes. We say that φ is a quotient map if φ is faithfully flat.

Thus φ is a quotient map if and only if φ is flat and surjective.

3.3 The Isomorphism Theorems for Algebraic Groups

Proposition 3.3.1

$\mathbf{AbSch}_k^{\text{ft}}$ is an abelian category.

Proposition 3.3.2

$\mathbf{AffAbSch}_k^{\text{ft}}$ is an abelian subcategory of $\mathbf{AbSch}_k^{\text{ft}}$.

4 Structure Theorem for Group Scheme

4.1 Chevalley's Structure Theorem

Theorem 4.1.1 (Chevalley's Structure Theorem)

Let k be a field. Let G be a smooth connected group scheme of finite type over k . Then G contains a largest smooth connected affine normal subgroup scheme N .

Theorem 4.1.2 (Barsotti-Chevalley Theorem)

Let k be a perfect field. Let G be a smooth connected group scheme of finite type over k . Let N be the largest smooth connected affine normal subgroup scheme. Then G/N is an abelian variety.

Relaxing the smoothness condition:

Theorem 4.1.3

Let k be a field. Let G be a smooth connected group scheme of finite type over k . Then there exists a connected affine normal subgroup scheme N of G such that G/N is an abelian variety.

4.2 The Anti-Affine Group Scheme

Definition 4.2.1 (The Associated Affine Group Scheme)

Let k be a field. Let G be a group scheme of finite type over k . Define the associated affine group scheme to be

$$G^{\text{aff}} = \text{Spec}(\mathcal{O}_G(G))$$

Proposition 4.2.2

Let k be a field. Let G be a group scheme of finite type over k . Then there is a natural isomorphism

$$\text{Hom}_{\text{GrpSch}_k}(G, H) \cong \text{Hom}_{\text{GrpSch}_k}(G^{\text{aff}}, H)$$

given by factorizing through the natural map $\phi : G \rightarrow G^{\text{aff}}$ for any affine group scheme H over k .

Proposition 4.2.3

Let k be a field. Let G be a group scheme of finite type over k . Then $\ker(G \rightarrow G^{\text{aff}})$ is an anti-affine group.

5 Diagonalizable Groups

5.1 The Group Algebra

Definition 5.1.1

Let k be a field. Define the functor

$$D : \mathbf{FGAb} \rightarrow \mathbf{GrpSch}_k^{\text{ft}}$$

by the composite

$$D = \text{Spec} \circ k[-]$$

where $k[-] : \mathbf{FGAb} \rightarrow \mathbf{FGCAlg}_k$ is the group algebra functor.

Lemma 5.1.2

Let k be a field. Let M be a finitely generated abelian group. Then we have

$$\widetilde{D(M)} = \text{Hom}_{\mathbf{Ab}}(M, (-)^\times)$$

By the fundamental theorem of finitely generated abelian groups, we have

$$\text{Spec}(k[M]) \cong \prod_{i=1}^k \mu_{n_i} \times \mathbb{G}_m^{\text{rank}(M)}$$

if

$$M \cong \prod_{i=1}^k \mathbb{Z}/n_i\mathbb{Z} \times \mathbb{Z}^{\text{rank}(M)}$$

Lemma 5.1.3

Let k be a field. Let M be a finitely generated abelian group. Then there is a group isomorphism

$$M \cong \chi(D(M))$$

where for each $A \in \mathbf{FGCAlg}_k$, the map

$$M \rightarrow \text{Hom}_{\mathbf{Ab}}(\text{Hom}_{\mathbf{Ab}}(M, A^\times), A^\times)$$

is given by $m \mapsto (f \mapsto f(m))$.

Proposition 5.1.4

Let k be a field such that $\text{char}(k) = p \neq 0$. Let M be a finitely generated abelian group. Then the following are true.

- $D(M)$ is connected if and only if the only torsion in G is p -torsion.
- $D(M)$ is smooth if and only if G has no p -torsion.
- $D(M)$ is smooth and connected if and only if G is free.

For characteristic 0, we already know that $D(M)$ is smooth by Cartier's theorem. For connectedness, we have:

Proposition 5.1.5

Let k be a field of characteristic 0. Let M be a finitely generated abelian group. Then $D(M)$ is connected if and only if M is free.

5.2 Diagonalizable Groups

Definition 5.2.1 (Diagonalizable Groups)

Let k be a field. Let G be an affine group scheme over k . We say that G is diagonalizable if

$$\mathcal{O}_G(G) = k\langle g \in G \mid g \text{ is a group-like element} \rangle$$

as k -vector spaces.

Definition 5.2.2 (The Category of Diagonalizable Group Schemes)

Let k be a field. Define the category $\mathbf{Diag}_k$ to be the full subcategory of $\mathbf{GrpSch}_k^{\text{ft}}$ that consists of diagonalizable group schemes.

Proposition 5.2.3

Let k be a field. Then the following are true.

- $D : \mathbf{FGAb} \rightarrow \mathbf{Diag}_k$ is an equivalence of categories, with inverse given by χ sending a group scheme to its abelian group of characters.
- D and χ are exact functors.

Corollary 5.2.4

Let k be a field. Let G be a diagonalizable affine group scheme over k . If $H \leq G$ is a subgroup of G , then H and G/H are diagonalizable.

5.3 Diagonalizable Representations

Definition 5.3.1 (Diagonalizable Representations)

Let k be a field. Let G be a group scheme of finite type over k . Let $\rho : G \rightarrow GL(V)$ be a representation of G . We say that ρ is diagonalizable if ρ is the sum of 1-dimensional representations.

Proposition 5.3.2

Let k be a field. Let G be a group scheme of finite type over k . Then the following are equivalent.

- G is diagonalizable.
- Every finite dimensional representation of G is diagonalizable.
- Every representation of G is diagonalizable.
- For every representation $\rho : G \rightarrow GL(V)$ of G , there is a decomposition

$$V = \bigoplus_{\chi \in \chi(G)} V_\chi$$

5.4 Tori

Definition 5.4.1 (Tori)

Let k be a field. Let G be a group scheme of finite type over k . We say that G is a tori if there exists a field extension F/k such that there is an isomorphism

$$G(F) \cong \mathbb{G}_m^n$$

for some $n \in \mathbb{N}^+$.

Definition 5.4.2 (Split Tori)

Let k be a field. Let G be a group scheme of finite type over k . We say that G is a split tori if there is an isomorphism

$$G \cong \mathbb{G}_m^n$$

for some $n \in \mathbb{N}^+$.

The split tori are precisely the smooth connected diagonalizable group schemes of finite type over k . They correspond to free finitely generated free abelian groups under the equivalence between D and χ .

Proposition 5.4.3

Let k be a field. Let T be a split tori. Then the following are true.

- Let G be a subgroup scheme of T . Then G is diagonalizable.
- Let G be a group scheme of finite type over k . Let $T \rightarrow G$ be a quotient map. Then G is a split tori.

Proposition 5.4.4

Let k be a field. Let T be a split torus over k . Then the following are true.

- Let T' be a split torus over k . Let $\phi : T \rightarrow T'$ be a morphism of group schemes. Then $\text{im}(\phi)$ is closed torus in T' .
- Let $H \leq T$ be a subgroup variety of T . If H is irreducible, then H is a torus.
- Let $H \leq T$ be a subgroup of T . Then T/H is a torus.

6 Examples of Group Schemes

6.1 The Affine Group Scheme of a Group Ring

Let R be a commutative ring. Let G be an abelian group. Recall that $R[G]$ is the group algebra whose elements are R -linear combination of elements of G and multiplication is induced by the binary operation of G .

Definition 6.1.1 (Group Scheme Associated to Group Algebra)

Let k be a field. Let G be an abelian group. Define the affine group scheme associated to the group algebra of G to be $\text{Spec}(k[G])$.

Definition 6.1.2 (Functor Associated to Group Algebra)

Let k be a field. Let G be an abelian group. Define the functor

$$D(G) : \mathbf{FGAlg}_k \rightarrow \mathbf{Grp}$$

as follows.

- For each k -algebra R , define $D(G)(R) = \text{Hom}_{\mathbf{Grp}}(G, R^\times)$.
- For each k -algebra homomorphism, $\varphi : R \rightarrow S$ is sent to the group homomorphism $\text{Hom}_{\mathbf{Grp}}(G, R^\times) \rightarrow \text{Hom}_{\mathbf{Grp}}(G, S^\times)$ which is post composition with φ .

Lemma 6.1.3

Let k be a field. Let G be an abelian group. Then the functor $D(G)$ represents the affine group scheme $\text{Spec}(k[G])$.

Proposition 6.1.4

Let k be a field. Let G be a finitely generated abelian group. Suppose that $\varphi : G \rightarrow \bigoplus_{i=1}^r \mathbb{Z}/b_i\mathbb{Z} \oplus \mathbb{Z}^n$ is a group isomorphism. Then φ induces an isomorphism of group schemes

$$\text{Spec}(k[G]) \cong \prod_{i=1}^r \mu_{b_i} \times \mathbb{G}_m^n$$

7 Finite Group Schemes and Cartier Duality

8 Group Varieties

All group schemes of finite type over a field are separated. So being a group variety only adds reducedness to the assumption. There are also the following facts.

- A group variety is irreducible if and only if it is connected if and only if it is geometrically connected.
- A group variety is reduced if and only if it is geometrically reduced.
- Over a perfect field (finite fields, characteristic 0, algebraically closed), all group varieties are smooth.
- Complete group varieties are commutative.
- Complete connected group varieties (i.e. abelian varieties) are anti-affine.
- Chevalley's structure theorem states that for an irreducible group variety G , there is an exact sequence

$$0 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 0$$

where N is affine and Q is an abelian variety.

In abelian varieties, we will see:

abelian varieties are projective.